



# New regression formula for toric intraocular lens calculations

Adi Abulafia, MD, Douglas D. Koch, MD, Li Wang, MD, PhD, Warren E. Hill, MD, Ehud I. Assia, MD, Maria Franchina, MD, Graham D. Barrett, MD

**PURPOSE:** To evaluate and compare the accuracy of 2 toric intraocular lens (IOL) calculators with or without a new regression formula.

**SETTING:** Ein-Tal Eye Center, Tel-Aviv, Israel, and the Lions Eye Institute, Nedlands, Western Australia, Australia.

**DESIGN:** Retrospective case series.

**METHODS:** A new regression formula (Abulafia-Koch) was developed to calculate the estimated total corneal astigmatism based on standard keratometry measurements. The error in the predicted residual astigmatism was calculated by the Alcon and Holladay toric IOL calculators with and without adjustments by the Abulafia-Koch formula. These results were compared with those of the Barrett toric calculator.

**RESULTS:** Data from 78 eyes were evaluated to validate the Abulafia-Koch formula. The centroid errors in predicted residual astigmatism were against-the-rule with the Alcon (0.55 diopter [D]) and Holladay (0.54 D) toric calculators and decreased to 0.05 D ( $P < .001$  [x-axis],  $P = .776$  [y-axis]) and 0.04 D ( $P < .001$  [x-axis],  $P = .726$  [y-axis]) with adjustments by the Abulafia-Koch formula. The Alcon and the Holladay toric calculators had a higher proportion of eyes within  $\pm 0.50$  D of the predicted residual astigmatism with the Abulafia-Koch formula (76.9% and 78.2%, respectively) than without it (both 30.8%). There were no significant differences between the results of the Abulafia-Koch-modified Alcon and the Holladay toric calculators and those of the Barrett toric calculator.

**CONCLUSION:** Adjustment of commercial toric IOL calculators by the Abulafia-Koch formula significantly improved the prediction of postoperative astigmatic outcome.

**Financial Disclosure:** Dr. Abulafia received a speaker's fee from Haag-Streit AG. Dr. Barrett has licensed the Barrett Toric Calculator to Haag-Streit AG. Dr. Koch is a consultant to Alcon Laboratories, Inc., Abbott Medical Optics, Inc., and Revision Optics, Inc. Dr. Hill is a paid consultant to Haag-Streit AG and Alcon Laboratories, Inc. None of the other authors has a financial or proprietary interest in any material or method mentioned.

*J Cataract Refract Surg 2016; 42:663–671 © 2016 ASCRS and ESCRS*

Toric intraocular lenses (IOLs) have improved the refractive outcome of patients with significant preexisting astigmatism who have cataract surgery.<sup>1–4</sup> However, the results are not always predictable.<sup>3,5,6</sup>

Corneal surgically induced astigmatism (SIA),<sup>2,7</sup> errors in the alignment of toric IOLs,<sup>8</sup> errors in measuring corneal astigmatism,<sup>9–12</sup> and errors in the method used to predict the required toric IOL and residual astigmatism after surgery<sup>12,13</sup> have been described as factors that might contribute to unexpected residual

astigmatism. Accurate measurement of net corneal astigmatism in cataract patients is crucial to achieve optimum postoperative uncorrected visual acuity and patient satisfaction.<sup>6,9,13–15</sup> Standard keratometry and Placido-disk corneal topographers that are based on anterior corneal measurement have been traditionally used to measure corneal astigmatism for determining the correct axis and cylinder power needed for a toric IOL<sup>4,16,17</sup> based on the assumption that the posterior cornea induces minimal refractive astigmatism and

can therefore be ignored in astigmatic calculations. However, several studies recently reported that methods of calculation that account for posterior corneal astigmatism might be more appropriate for toric IOL calculations. Those methods apply nomograms,<sup>13</sup> coefficients of adjustment,<sup>6</sup> direct measurements of the posterior cornea,<sup>10,18</sup> a ray-tracing approach using combined keratometry and tomography data,<sup>14</sup> intraoperative aberrometry,<sup>19</sup> and mathematical models.<sup>12</sup>

The purpose of this study was to describe a new regression formula that can calculate the estimated total corneal astigmatism using standard keratometry (K) measurements and to compare the error in the predicted residual astigmatism of 2 representative toric IOL calculators with and without the use of the new formula. Those results were further compared with those of the Barrett toric calculator.<sup>A</sup>

## PATIENTS AND METHODS

### Patients

Institutional review board approval was obtained for this study. The study reviewed case records of consecutive patients who had cataract extraction with Acrysof toric IOL (models SN6AT3 to SN6AT9, Alcon Laboratories) implantation through a 2.4 mm clear corneal incision (CCI) by multiple surgeons at the Ein-Tal Eye Center, a private practice in Tel-Aviv, Israel, from February 2011 to September 2012.

Inclusion criteria were uneventful cataract surgery, a manifest refraction at least 1 month postoperatively with a corrected distance visual acuity of 20/30 or better, postoperative measurements at least 1 month after surgery (same day as for the manifest refraction) with optical low-coherence reflectometry (OLCR) (Lenstar LS 900, Haag-Streit AG) compatible with strict validation criteria by Hill,<sup>B</sup> and postoperative toric IOL axis alignment measured by slitlamp and photography. All measurements were recorded at the

same postoperative visit. The exclusion criteria were previous ocular trauma or surgery, irregular corneal astigmatism (eg, keratoconus), intraoperative or postoperative complications, contact lens wear, and postoperative toric IOL decentration or tilt under slitlamp examination.

### Evaluation of Manifest Cylinder

The evaluation of the manifest cylinder was performed in a technique similar to that described by Hoffmann et al.<sup>11</sup> After the assessment of the uncorrected distance visual acuity (UDVA) and the best spherical correction, a cross-cylinder was placed in front of the eye in the 0-degree, 90-degree, 45-degree, and 135-degree axes. Its power was chosen according to the visual acuity. For a UDVA worse than 20/40, a  $\pm 0.5$  cross cylinder was used; from 20/40 upward, a  $\pm 0.25$  cross cylinder was used. If the patient reported no improvement, a manifest cylinder of zero was recorded. If a cylinder in 1 of the principal meridians was accepted, its axis was fine-tuned by quickly turning the cross cylinder using the "handle on axis" approach. After the axis was determined, the required cylinder power was assessed by quickly turning the cross cylinder axis-on-axis and adjusting the refractive cylinder accordingly.

### Abulafia-Koch Formula Development

The estimated total corneal astigmatism was calculated for each eye by subtracting the toric IOL power at the corneal plane, which was calculated using the Holladay 1 formula<sup>20</sup> with the method described by Fam and Lim,<sup>21</sup> from the cylinder power of the postoperative manifest refraction adjusted to the corneal plane. A linear regression was derived between the estimated net corneal astigmatism and the astigmatism measured by the OLCR device, separate for the  $x$  component ( $y = 0.508 + 0.926 \times x$ ) and  $y$  component ( $y = 0.009 + 0.932 \times x$ ). To calculate the predicted total corneal astigmatism from the OLCR device's anterior corneal measurements, the measured corneal astigmatism was first separated into its  $x$  and  $y$  components<sup>22</sup>; then, the new regression formulas were applied for each component to derive new  $x'$  and  $y'$  values. A new vector (the estimated net corneal astigmatism) was then reconstructed using the  $x'$  and  $y'$  values.<sup>22</sup> Vector analysis was used in all calculations.<sup>22</sup>

### Abulafia-Koch Formula Validation

A validation dataset was used to evaluate the accuracy of the Abulafia-Koch formula. The dataset comprised consecutive cases with toric Acrysof IOL (models SN6AT2 to SN6AT7) implantation through a 2.4 mm CCI by the same surgeon (G.D.B.) from another center (Lion's Eye Institute, private practice, Nedlands, Western Australia) from January 2012 to October 2014. The same inclusion and exclusion criteria as above were followed with the exception that the postoperative measurements by partial coherence interferometry (PCI) (IOLMaster 500, Carl Zeiss Meditec AG) at least 1 month after surgery were used instead of data from the OLCR device.

### Toric Intraocular Lens Calculators

The Alcon toric calculator model<sup>C</sup> uses a constant ratio as the basis to determine the effect of the toric cylinder power at the corneal plane. For the Holladay toric calculator, the

Submitted: November 3, 2015.

Final revision submitted: January 29, 2016.

Accepted: February 1, 2016.

From Ein-Tal Eye Center (Abulafia, Assia), Tel-Aviv, Assaf Harofeh Medical Center (Abulafia), Zerifin, Meir Medical Center (Assia), and Kfar-Saba, affiliated with Tel-Aviv University, Tel-Aviv, Israel; Cullen Eye Institute (Koch, Wang), Baylor College of Medicine, Houston, Texas, and East Valley Ophthalmology (Hill), Mesa, Arizona, USA; Sir Charles Gairdner Hospital (Franchina, Barrett) and Lions Eye Institute (Barrett), Nedlands, Western Australia, Australia.

Guy Kleinmann, MD, Arie L. Marcovich, MD, PhD, and Adi Michaeli, MD, were physicians and surgeons for some of Ein-Tal's patients. Shay Ofir, MD, Adi Levy, BSc, and Olga Reitblat, MD, helped with the data acquisition for Ein-Tal's patients.

Corresponding author: Adi Abulafia, MD, Zerifin 70300, Israel. E-mail: [adi.abulafia@gmail.com](mailto:adi.abulafia@gmail.com).

method described by Fam and Lim<sup>21</sup> was used in which the IOL spherical power and the predicted effective lens position (ELP) in each eye are used to estimate the effective IOL cylinder power at the corneal plane. The Barrett Toric calculator<sup>12,A</sup> uses the Universal II formula<sup>23</sup> to calculate an ELP, which is also influenced by the spherical equivalent and cylindrical toric IOL power. This calculator predicts posterior corneal astigmatism based on a theoretical model and was recently modified to include a regression component.

## Calculation Methods

The results derived from the Alcon and the Holladay 1 toric calculator models were evaluated using the PCI device K measurements with and without the Abulafia-Koch formula. The online Barrett toric calculator was evaluated using the PCI device K measurements without adjustment.

## Evaluation of Error in Predicted Residual Astigmatism

Astigmatism values were calculated by subtracting the flat K values from the steep K values. The assumed toric IOL cylinder powers at the corneal plane were derived for the Alcon and Holladay 1 models and for the Barrett toric calculator. To achieve optimum accuracy, the postoperative-measured toric IOL alignment was used rather than the planned toric IOL alignment. The predicted residual astigmatism at the corneal plane was calculated by the sum of the assumed toric IOL cylinder power at the corneal plane and the corneal astigmatism measured by each device. The error in predicted residual astigmatism was calculated by subtracting the predicted residual astigmatism at the corneal plane from the postoperative subjective refraction referenced to the corneal plane. Vector analysis was used in all calculations.<sup>22</sup> The mean absolute error, median absolute error, and centroid error in the predicted residual astigmatism were calculated for each method.

## Statistical Analysis

A sample-size calculation was performed to detect an astigmatic prediction error of 0.125 diopter (D) and a standard deviation of 0.30 D. The calculation showed that 63 eyes were required to obtain a significance level ( $\alpha$ ) of 0.05 and a test power of 0.90. Comparisons were performed for astigmatic predicted errors, including the  $x$ -axis and  $y$ -axis components for each toric calculator and method of adjustment. Data were checked for normality by Shapiro-Wilks tests. The Wilcoxon nonparametric test was used for comparisons of 2 samples. Multiple pairwise comparisons were performed using the Dunn procedure with Bonferroni correction. Differences were considered statistically significant according to the Bonferroni-corrected significance level for each comparison. For the double-angle plot diagrams, 95% confidence intervals were constructed according to formal bivariate statistical principles.<sup>24</sup> Statistical analyses were performed with the Xlstat software (version 2015.3.01.19199, Addinsoft).

## RESULTS

In the main dataset used to develop the Abulafia-Koch formula, 68 eyes of 48 patients met the inclusion criteria. The mean age of the 17 women and 31 men was

66 years  $\pm$  9.5 (SD) (range 42 to 86 years). The mean axial length (AL) was  $24.88 \pm 1.48$  mm (range 22.27 to 28.02 mm), and the mean IOL power was  $17.6 \pm 4.5$  D (range 7.0 to 26.0 D). Forty-three eyes (63.2%) had with-the-rule (WTR) anterior corneal astigmatism (60 to 120 degrees), 22 eyes (32.4%) had against-the-rule (ATR) anterior corneal astigmatism (0 to 30 degrees or 150 to 180 degrees), and 3 eyes (4.4%) had oblique anterior corneal astigmatism (30 to 60 degrees or 120 to 150 degrees). The mean anterior corneal astigmatism was  $2.34 \pm 0.85$  D (range 1.14 to 4.44 D).

In the validation dataset from the center in Australia, 78 eyes of 61 patients met the inclusion criteria. The mean age of the 36 women and 25 men was  $70 \pm 9.0$  years (range 45 to 83 years). The mean AL was  $23.78 \pm 1.30$  mm (range 20.99 to 26.99 mm), and the mean IOL power was  $21.3 \pm 4.4$  D (range 8.5 to 31.0 D). Twenty-two eyes (28.2%) had WTR anterior corneal astigmatism, 45 eyes (57.7%) had ATR anterior corneal astigmatism, and 11 eyes (14.1%) had oblique anterior corneal astigmatism. The mean anterior corneal astigmatism was  $1.30 \pm 0.92$  D (range 0.21 to 4.37 D).

Table 1 shows the characteristics of the IOLs that were implanted.

## Abulafia-Koch Formula Development

A high correlation was found between the  $x$  component and  $y$  component of the estimated net corneal astigmatism and the astigmatism measured by the OLCR device (all  $P < .0001$ ), with  $R^2$  values of 0.986 and 0.963, respectively (Figure 1).

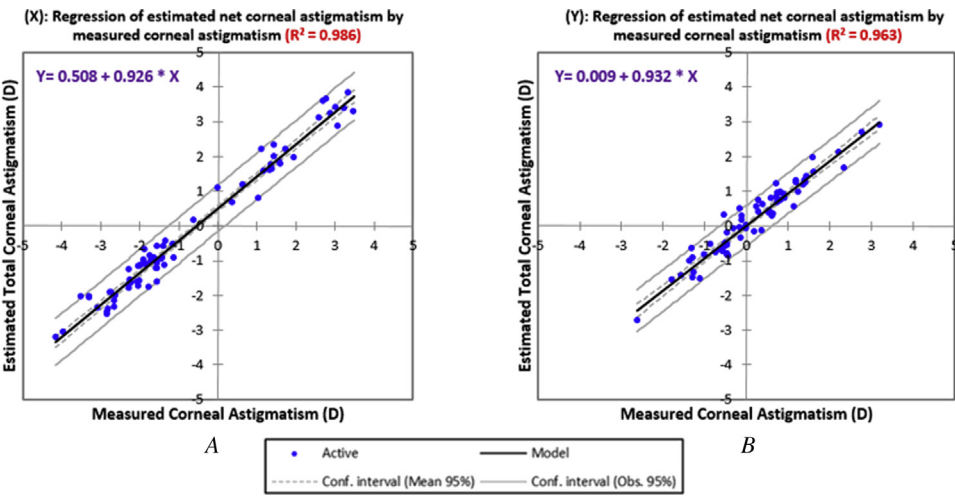
## Absolute Error in Predicted Residual Astigmatism

Table 2 and Figure 2 show the median absolute errors in predicted residual astigmatism. The median

**Table 1.** Type of IOL and frequency of implantation.

| IOL Model | Number (%)                     |                                |
|-----------|--------------------------------|--------------------------------|
|           | Dataset for Developing Formula | Dataset for Validating Formula |
| SN6AT2    | 0                              | 17 (21.8)                      |
| SN6AT3    | 1 (1.5)                        | 39 (50.0)                      |
| SN6AT4    | 20 (29.4)                      | 13 (16.7)                      |
| SN6AT5    | 17 (25.0)                      | 3 (3.8)                        |
| SN6AT6    | 11 (16.2)                      | 1 (1.3)                        |
| SN6AT7    | 11 (16.2)                      | 5 (6.4)                        |
| SN6AT8    | 3 (4.4)                        | 0                              |
| SN6AT9    | 5 (7.4)                        | 0                              |

IOL = intraocular lens



**Figure 1.** Correlation of the estimated total corneal astigmatism and the measured astigmatism (OLCR device) for the x (A) and y (B) vector components (both  $P < .001$ ) (Conf. = confidence).

absolute errors in predicted residual astigmatism were not statistically significantly different between the Alcon and the Holladay toric calculator models ( $P = .126$ ). Application of the Abulafia-Koch formula reduced the median absolute error for the Alcon and the Holladay toric calculators (both  $P < .001$ ). There was no significant difference between the Alcon and the Holladay toric calculator models adjusted by the Abulafia-Koch formula ( $P = .324$ ). The median absolute error in predicted residual astigmatism with the Barrett toric calculator was significantly lower than with the Alcon and the Holladay toric calculator models with no adjustments (both  $P < .001$ ) but not significantly different when the Abulafia-Koch formula was applied ( $P = .088$  and  $P = .331$ , respectively).

Table 3 show the proportion of eyes that had an absolute error in predicted residual astigmatism equal to or less than 0.50 D, 0.75 D, and 1.00 D. A higher proportion of eyes had an absolute error in predicted residual astigmatism of equal to or less than 0.50 D, 0.75 D, and 1.00 D when the Abulafia-Koch formula was used to adjust the Alcon and Holladay toric calculator models compared with no adjustments. Similar

proportions of eyes were found with the Barrett toric calculator compared with the Alcon and Holladay toric calculators with Abulafia-Koch adjustments.

Centroid Error in Predicted Residual Astigmatism

The centroid errors in predicted residual astigmatism were not significantly different between Alcon and Holladay toric calculators ( $P = .872$  [x-axis],  $P = .912$  [y-axis]) (Table 2 and Figure 3). Applying the Abulafia-Koch formula significantly reduced the centroid errors in predicted residual astigmatism to 0.05 D ( $P < .001$  [x-axis],  $P = .776$  [y-axis]) and 0.04 D ( $P < .001$  [x-axis],  $P = .726$  [y-axis]), respectively. No significant difference was noted between the Alcon and the Holladay toric calculator models after adjustment by the Abulafia-Koch formula ( $P = .880$  [x-axis],  $P = .965$  [y-axis]). The Barrett toric calculator centroid error in predicted residual astigmatism was significantly lower than the Alcon and the Holladay calculator models with no Abulafia-Koch adjustments ( $P < .001$  [x-axis],  $P = .834$  [y-axis] and  $P < .001$  [x-axis],  $P = .750$  [y-axis], respectively) but not significantly different from the Alcon and Holladay

| Value                        | Method of Calculation  |                                                  |                           |                                                     |                          |
|------------------------------|------------------------|--------------------------------------------------|---------------------------|-----------------------------------------------------|--------------------------|
|                              | Alcon Toric Calculator | Alcon Toric Calculator (+ Abulafia-Koch Formula) | Holladay Toric Calculator | Holladay Toric Calculator (+ Abulafia-Koch Formula) | Barrett Toric Calculator |
| Mean $\pm$ SD (D)            | 0.67 $\pm$ 0.28        | 0.38 $\pm$ 0.22                                  | 0.66 $\pm$ 0.27           | 0.38 $\pm$ 0.22                                     | 0.37 $\pm$ 0.21          |
| Range (D)                    | 0.10, 1.41             | 0.06, 1.21                                       | 0.14, 1.29                | 0.08, 1.13                                          | 0.05, 1.06               |
| Median (D)                   | 0.66                   | 0.34                                             | 0.68                      | 0.34                                                | 0.36                     |
| Mean centroid (D@°) $\pm$ SD | 0.55 $\pm$ 0.33@180    | 0.05 $\pm$ 0.31@174                              | 0.54 $\pm$ 0.33@1         | 0.04 $\pm$ 0.31@176                                 | 0.05 $\pm$ 0.30@176      |

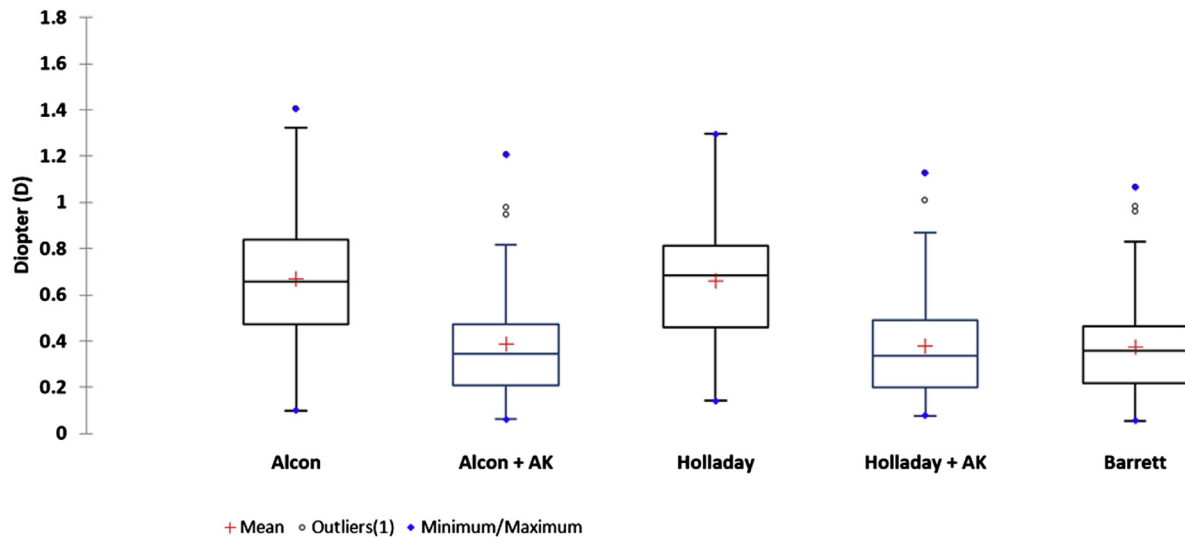


Figure 2. Absolute error in predicted residual astigmatism by method of calculation, all eyes ( $n = 78$ ) (AK = Abulafia-Koch formula).

calculator models when using Abulafia-Koch adjustments ( $P = .900$  [ $x$ -axis],  $P = .940$  [ $y$ -axis], and  $P = .980$  [ $x$ -axis],  $P = .975$  [ $y$ -axis], respectively).

## DISCUSSION

In general, the use of toric IOLs has greatly improved the refractive outcomes of cataract surgery in patients with preexisting astigmatism. Nevertheless, unexpected residual astigmatism can occur after their implantation. One cause of this untoward occurrence is inaccurate estimation of the total corneal astigmatism.<sup>6,9,11-13</sup> A perfect method to calculate preoperative total corneal astigmatism has not been established.<sup>25</sup> To our knowledge, none of the manufacturers' online toric IOL calculators account for posterior corneal astigmatism,<sup>6</sup> resulting in ATR prediction errors when using radii measurements based on the anterior corneal surface alone.<sup>12,13</sup> The Barrett toric calculator,<sup>A</sup> which uses a mathematic model to estimate posterior corneal astigmatism, was shown to be better than standard toric calculators

that use anterior cornea-based keratometry<sup>12</sup> and those that use direct measurements of the posterior cornea by the Scheimpflug camera device (Pentacam, Oculus Optikgeräte GmbH).<sup>26</sup>

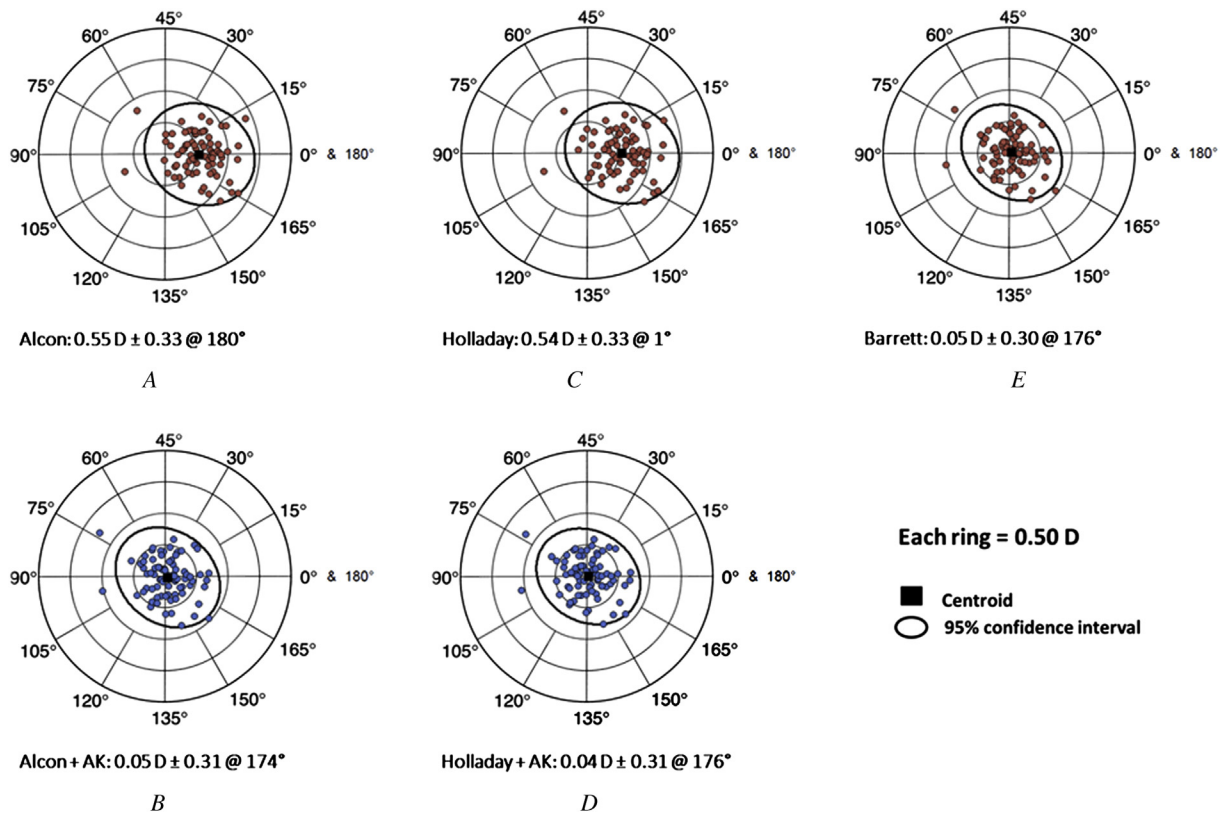
The aim of our present study was to describe a new regression formula (Abulafia-Koch) that calculates an estimated net corneal astigmatism using standard K measurements (based on the anterior cornea). We compared the error in predicted residual astigmatism after toric IOL implantation using 2 representative commercial toric IOL calculator models, with and without applying the Abulafia-Koch formula. These results were further compared with the Barrett toric calculator. To avoid the noise associated with variable corneal SIA and incorrect toric IOL alignment, measurements and calculations were performed using postoperative K values and postoperative toric IOL axis alignment values. Our results were derived by calculating the absolute and centroid astigmatic prediction errors.

The  $x$  and  $y$  components of the estimated total corneal astigmatism and the measured astigmatism by the

Table 3. Proportion of eyes with absolute error in predicted residual astigmatism below or equal to 0.50 D, 0.75 D and 1.00 D (all 78 eyes).

| Method of Calculation          | Proportion          |                     |                     |
|--------------------------------|---------------------|---------------------|---------------------|
|                                | Within $\pm 0.50$ D | Within $\pm 0.75$ D | Within $\pm 1.00$ D |
| Alcon (PCI)                    | 30.8                | 60.3                | 87.2                |
| Alcon (PCI) + Abulafia-Koch    | 76.9                | 93.6                | 98.7                |
| Holladay (PCI)                 | 30.8                | 61.5                | 88.5                |
| Holladay (PCI) + Abulafia-Koch | 78.2                | 93.6                | 97.4                |
| Barrett (PCI)                  | 79.5                | 94.9                | 98.7                |

PCI = partial coherence interferometry device



**Figure 3.** Double-angle plots of errors in predicted residual astigmatism, all eyes ( $n = 78$ ). The Alcon toric calculator with no adjustments (A) and adjusted by the Abulafia-Koch formula (B). The Holladay toric calculator with no adjustments (C) and adjusted by the Abulafia-Koch formula (D). The Barrett toric calculator with no adjustments (E) (AK = Abulafia-Koch formula).

Lenstar OLCR device were highly correlated (all  $P < .001$ ), with  $R^2$  values of 0.986 and 0.963, respectively. This finding is compatible with the relation between the anterior and posterior corneas first described in 1890 by Javal<sup>27</sup> (Javal's rule), which was later simplified in 1988 by Grosvenor et al.<sup>28</sup> and confirmed by several authors for diverse populations<sup>29–33</sup> as well as for pseudophakic eyes.<sup>34,35</sup> The results of our regression analysis indicate that the  $x$  component (estimated net corneal astigmatism =  $0.508 + 0.926 \times$  measured corneal astigmatism) plays a pivotal role by expressing a rather predicted pattern of deviation between the estimated net and the measured corneal astigmatism, whereas the  $y$  component role is almost negligible (estimated net corneal astigmatism =  $0.009 + 0.932 \times$  measured corneal astigmatism). These findings correlate well with the finding by Koch et al.,<sup>9</sup> who reported a steep posterior corneal meridian aligned vertically (WTR) in 86.6% of 715 corneas of 435 consecutive cases, as well as with findings in other studies.<sup>11,18</sup> Because the posterior cornea acts as a negative lens, the effect of the posterior cornea will cause anterior corneal measurements to underestimate the total corneal astigmatism on the horizontal meridian. Thus, using anterior corneal measurements for toric IOL calculations tends to lead to ATR prediction errors of the

postoperative residual astigmatism, which could result in overcorrection of eyes with WTR astigmatism and undercorrection of eyes with ATR astigmatism.<sup>6,9–11</sup>

The double-angle plots diagrams in Figure 3, A and C, show the deviation of the centroid errors in the predicted residual astigmatism away from the 90-degree meridian toward the 180-degree meridian along the  $x$ -axis, whereas the centroid errors over the  $y$ -axis remain almost zero. This deviation along the  $x$ -axis is addressed by the Abulafia-Koch formula, as seen in Figure 3, B and D. Indeed, the use of the Alcon and the Holladay toric calculator models with standard K measurements (IOLMaster) led to ATR centroid errors of more than 0.50 D and to median absolute errors in the predicted residual astigmatism of 0.66 D and 0.68 D, respectively, with only 30.8% of eyes being within  $\pm 0.50$  D of the absolute errors in predicted residual astigmatism. These results correlate with those in previous studies<sup>12,13,26</sup> and indicate that posterior corneal astigmatism should be taken into consideration when performing toric IOL calculations. In contrast, adjustment of the same K measurements using the Abulafia-Koch formula decreased the centroid errors in predicted residual astigmatism to 0.05 D and 0.04 D, respectively (both  $P < .001$ ,  $x$ -axis), and the

median absolute errors to 0.34 D (both  $P < .001$ ), with 76.9% and 78.2% of eyes, respectively, being within  $\pm 0.50$  D of the absolute error in predicted residual astigmatism. These results were similar to those obtained by the Barrett toric calculator without adjustments.

The Alcon toric calculator model<sup>C</sup> uses a fixed ratio to calculate the toric effect of the IOL at the corneal plane, whereas the use of Fam and Lim's<sup>21</sup> meridional method of calculating effective IOL toricity with the Holladay 1 formula<sup>20</sup> theoretically should be more accurate because it incorporates IOL spherical power and the estimated ELP for this calculation. However, we found no significant differences between the 2 calculators with or without the use of the Abulafia-Koch formula, suggesting that the advantage of the Holladay 1 calculator model might be limited in ordinary eyes. Nevertheless, the use of a toric calculator that takes into account the ELP and IOL power is beneficial for eyes with unusual ELP and/or extreme IOL/toric powers.<sup>D</sup>

The standard calculators with the Abulafia-Koch formula adjustments and the Barrett toric calculator yielded similar results in predicting the residual corneal astigmatism using anterior cornea-based K measurements. The Barrett toric calculator, by implementing the Barrett Universal formula II,<sup>E</sup> has an integral solution, calculating both IOL spherical power and toric power at the corneal plane, whereas the Abulafia-Koch formula solely adjusts standard K measurements to the estimated net corneal power, thereby requiring independent calculation of IOL spherical power but facilitating the formula's use with any toric IOL calculator. As opposed to nomograms<sup>13</sup> or a coefficient of adjustment,<sup>6</sup> both solutions calculate the estimated net corneal astigmatism as a vector, thereby internally addressing the changes in the magnitude and meridian for WTR, ATR, and oblique astigmatism.

One limitation of our study is that we did not use direct measurements of the net corneal astigmatism from Scheimpflug camera-based devices, optical coherence tomography, or new devices, such as the Cassini corneal shape analyzer (i-Optics Corp.). We also did not use intraoperative aberrometry<sup>19</sup> or a ray-tracing method, such as the one suggested by Hoffmann et al.<sup>14</sup> Additional studies comparing the Abulafia-Koch formula adjustments and the Barrett IOL calculator to intraoperative aberrometry and to devices that directly measure the net corneal astigmatism by a ray-tracing approach are warranted. An additional limitation of our study is that we validated the Abulafia-Koch formula using IOLMaster measurements. Although applying the Abulafia-Koch formula was more accurate than using the toric IOL calculator models without it, Lenstar measurements would

probably better emphasize the differences between these 2 groups.<sup>12</sup> Furthermore, when we derived the Abulafia-Koch regression formula (using the same 68 eyes from the center 1 dataset) based on the IOLMaster measurements, it yielded inferior prediction results compared with the regression that was derived based on Lenstar measurements (data not shown). Finally, we did not examine the application of the Abulafia-Koch formula in a prospective clinical setting; rather, we recalculated the error in predicted residual astigmatism had the new formula been used for eyes that had toric IOL implantation, omitting the influence of the SIA, IOL rotation, and individual surgeon's input (including the use of primary and secondary devices for validation of the corneal astigmatism measurements, as suggested by Hill<sup>F</sup>). Additional studies that evaluate the actual refractive outcome of clinical application of the formula are warranted.

In conclusion, anterior cornea-based K measurements adjusted by the Abulafia-Koch formula significantly reduced the errors in predicted residual astigmatism in the Alcon and the Holladay toric calculator models. These results were similar to those obtained by the Barrett toric calculator with no adjustments. The Abulafia-Koch formula can be easily integrated into any toric calculator as an add-on option to improve toric IOL calculation accuracy.

#### WHAT WAS KNOWN

- Devices that measure the anterior cornea with standard toric IOL calculators tend to result in undercorrection of eyes with ATR astigmatism and overcorrection of eyes with WTR astigmatism.
- The Barrett toric calculator significantly reduces errors in prediction of residual astigmatism in toric IOL calculations.

#### WHAT THIS PAPER ADDS

- Standard toric IOL calculators with anterior cornea-based keratometry adjusted by the new Abulafia-Koch formula had significantly reduced errors in the prediction of residual astigmatism in toric IOL calculations to a level similar to that of the Barrett toric calculator without adjustments.

#### REFERENCES

1. Holland E, Lane S, Horn JD, Ernest P, Arleo R, Miller KM. T The AcrySof toric intraocular lens in subjects with cataracts and corneal astigmatism; a randomized, subject-masked, parallel-group, 1-year study. *Ophthalmology* 2010; 117:2104–2211
2. Goggin M, Moore S, Esterman A. Toric intraocular lens outcome using the manufacturer's prediction of corneal plane equivalent intraocular lens cylinder power. *Arch Ophthalmol* 2011; 129:1004–1008. Available at: <http://archophth.jamanetwork.com/article.aspx?articleid=1106788>. Accessed February 21, 2016

3. Mendicute J, Irigoyen C, Aramberri J, Ondarra A, Montés-Micó R. Foldable toric intraocular lens for astigmatism correction in cataract patients. *J Cataract Refract Surg* 2008; 34:601–607
4. Gundersen KG, Masket S, Osher RH, Snyder ME, Vann RR, Solomon KD, Hill WE. Prospective multicenter study of toric IOL outcomes when dual zone automated keratometry is used for astigmatism planning. *J Refract Surg* 2013; 29:804–809
5. Sun X-Y, Vicary D, Montgomery P, Griffiths M. Toric intraocular lenses for correcting astigmatism in 130 eyes. *Ophthalmology* 2000; 107:1776–1781; discussion by RM Kershner, 1781–1782
6. Goggin M, Zamora-Alejo K, Esterman A, van Zyl L. Adjustment of anterior corneal astigmatism values to incorporate the likely effect of posterior corneal curvature for toric intraocular lens calculation. *J Refract Surg* 2015; 31:98–102
7. Hill W, Potvin R. Monte Carlo simulation of expected outcomes with the AcrySof® toric intraocular lens. *BMC Ophthalmol* 2008; 8:22. Available at: <http://www.ncbi.nlm.nih.gov/pmc/articles/PMC2586009/pdf/1471-2415-8-22.pdf>. Accessed February 21, 2016
8. Jin H, Limberger I-J, Ehmer A, Guo H, Auffarth GU. Impact of axis misalignment of toric intraocular lenses on the refractive outcomes after cataract surgery. *J Cataract Refract Surg* 2010; 36:2061–2072
9. Koch DD, Ali SF, Weikert MP, Shirayama M, Jenkins R, Wang L. Contribution of posterior corneal astigmatism to total corneal astigmatism. *J Cataract Refract Surg* 2012; 38:2080–2087
10. Savini G, Næser K. An analysis of the factors influencing the residual refractive astigmatism after cataract surgery with toric intraocular lenses. *Invest Ophthalmol Vis Sci* 2015; 56:827–835. Available at: <http://iovs.arvojournals.org/article.aspx?articleid=2212837>. Accessed February 21, 2016
11. Hoffmann PC, Abraham M, Hirschschall N, Findl O. Prediction of residual astigmatism after cataract surgery using swept source fourier domain optical coherence tomography. *Curr Eye Res* 2014; 39:1178–1186
12. Abulafia A, Barrett GD, Kleinmann G, Ofir S, Levy A, Marcovich AL, Michaeli A, Koch DD, Wang L, Assia EI. Prediction of refractive outcomes with toric intraocular lens implantation. *J Cataract Refract Surg* 2015; 41:936–944
13. Koch DD, Jenkins RB, Weikert MP, Yeu E, Wang L. Correcting astigmatism with toric intraocular lenses: effect of posterior corneal astigmatism. *J Cataract Refract Surg* 2013; 39:1803–1809
14. Hoffmann PC, Wahl J, Hütz WW, Preußner P-R. A ray tracing approach to calculate toric intraocular lenses. *J Refract Surg* 2013; 29:402–408
15. Savini G, Hoffer KJ, Ducoli P. A new slant on toric intraocular lens power calculation. *J Refract Surg* 2013; 29:348–354
16. Lee H, Chung JL, Kim EK, Sgrignoli B, Kim TI. Univariate and bivariate polar value analysis of corneal astigmatism measurements obtained with 6 instruments. *J Cataract Refract Surg* 2012; 38:1608–1615
17. Hill W, Osher R, Cooke D, Solomon K, Sandoval H, Salas-Cervantes R, Potvin R. Simulation of toric intraocular lens results: manual keratometry versus dual-zone automated keratometry from an integrated biometer. *J Cataract Refract Surg* 2011; 37:2181–2187
18. Preussner P-R, Hoffmann P, Wahl J. Impact of posterior corneal surface on toric intraocular lens (IOL) calculation. *Curr Eye Res* 2015; 40:809–814
19. Hatch KM, Woodcock EC, Talamo JH. Intraocular lens power selection and positioning with and without intraoperative aberrometry. *J Refract Surg* 2015; 31:237–242
20. Holladay JT, Prager TC, Chandler TY, Musgrove KH, Lewis JW, Ruiz RS. A three-part system for refining intraocular lens power calculations. *J Cataract Refract Surg* 1988; 14:17–24
21. Fam HB, Lim KL. Meridional analysis for calculating the expected spherocylindrical refraction in eyes with toric intraocular lenses. *J Cataract Refract Surg* 2007; 33:2072–2076
22. Holladay JT, Moran JR, Kezirian GM. Analysis of aggregate surgically induced refractive change, prediction error, and intraocular astigmatism. *J Cataract Refract Surg* 2001; 27:61–79
23. Barrett GD. An improved universal theoretical formula for intraocular lens power prediction. *J Cataract Refract Surg* 1993; 19:713–720
24. Naeser K, Knudsen EB, Hansen MK. Bivariate polar value analysis of surgically induced astigmatism. *J Refract Surg* 2002; 18:72–78
25. Lee H, Kim TI, Kim EK. Corneal astigmatism analysis for toric intraocular lens implantation: precise measurements for perfect correction. *Curr Opin Ophthalmol* 2015; 26:34–38
26. Abulafia A, Hill WE, Franchina M, Barrett GD. Comparison of methods to predict residual astigmatism after intraocular lens implantation. *J Refract Surg* 2015; 31:699–707
27. Javal É. *Mémoires d'Ophtalmométrie: Annotés et Précédés d'une Introduction*. Paris, France, G. Masson, 1890
28. Grosvenor T, Quintero S, Perrigin DM. Predicting refractive astigmatism: a suggested simplification of Javal's rule. *Am J Optom Physiol Opt* 1988; 65:292–297
29. Remón L, Benlloch J, Furlan WD. Corneal and refractive astigmatism in adults: a power vectors analysis. *Optom Vis Sci* 2009; 86:1182–1186. Available at: [http://journals.lww.com/optvissci/Fulltext/2009/10000/Corneal\\_and\\_Refractive\\_Astigmatism\\_in\\_Adults\\_\\_A.9.aspx](http://journals.lww.com/optvissci/Fulltext/2009/10000/Corneal_and_Refractive_Astigmatism_in_Adults__A.9.aspx). Accessed February 21, 2016
30. Tong L, Carkeet A, Saw S-M, Tan DTH. Corneal and refractive error astigmatism in Singaporean schoolchildren: a vector-based Javal's rule. *Optom Vis Sci* 2001; 78:881–887. Available at: [http://journals.lww.com/optvissci/Abstract/2001/12000/Corneal\\_and\\_Refractive\\_Error\\_Astigmatism\\_in.10.aspx](http://journals.lww.com/optvissci/Abstract/2001/12000/Corneal_and_Refractive_Error_Astigmatism_in.10.aspx). Accessed February 21, 2016
31. Alpíns NA. New method of targeting vectors to treat astigmatism. *J Cataract Refract Surg* 1997; 23:65–75
32. Auger P. Confirmation of the simplified Javal's rule [letter]. *Am J Optom Physiol Opt* 1988; 65:915. Available at: [http://journals.lww.com/optvissci/Citation/1988/11000/CONFIRMATION\\_OF\\_THE\\_SIMPLIFIED\\_JAVAL\\_S\\_RULE\\_.10.aspx](http://journals.lww.com/optvissci/Citation/1988/11000/CONFIRMATION_OF_THE_SIMPLIFIED_JAVAL_S_RULE_.10.aspx). Accessed February 21, 2016
33. Lam AKC, Chan CCY, Lee MH, Wong KM. The aging effect on corneal curvature and the validity of Javal's rule in Hong Kong Chinese. *Curr Eye Res* 1999; 18:83–90
34. Teus MA, Arruabarrena C, Hernández-Verdejo JL, Sales-Sanz A, Sales-Sanz M. Correlation between keratometric and refractive astigmatism in pseudophakic eyes. *J Cataract Refract Surg* 2010; 36:1671–1675
35. Bae JG, Kim SJ, Choi YI. Pseudophakic residual astigmatism. *Korean J Ophthalmol* 2004; 18:116–120. Available at: <http://ekjo.org/Synapse/Data/PDFData/0065KJO/kjo-18-116.pdf>. Accessed February 21, 2016

#### OTHER CITED MATERIAL

- A. Barrett toric calculator. Available at [http://www.apacrs.org/toric\\_calculator/Toric%20Calculator.aspx](http://www.apacrs.org/toric_calculator/Toric%20Calculator.aspx). Accessed February 21, 2016
- B. Hill WE, Abulafia A, Barrett GD, Kleinmann G, Assia EI. "Optimizing Toric IOL Outcome," instruction course given at the XXXIII Congress of the European Society of Cataract and Refractive Surgeons, Barcelona, Spain, September 2015
- C. Alcon Surgical, Inc. AcrySof® Toric IOL Web Based Calculators. Available at: <http://www.acrysoftoriccalculator.com>. Accessed February 21, 2016
- D. Holladay JT. Improving toric IOL outcomes, parts 1 and 2. *Ocular Surgery News* May 25 and June 10, 2011. Part 1

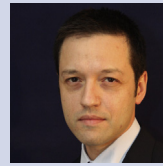
available at: <http://www.healio.com/ophthalmology/refractive-surgery/news/print/ocular-surgery-news/%7Ba8a94136-740c-4efd-8e5f-7ec59e2c0ad0%7D/improving-toric-iol-outcomes-part-1>.

Part 2 available at: <http://www.healio.com/ophthalmology/refractive-surgery/news/print/ocular-surgery-news/%7B61499f4b-dc6e-4e52-af42-e2dea31e39f0%7D/improving-toric-iol-outcomes-part-2>.

Accessed February 22, 2016

- E. Barrett GD. Barrett Universal II Formula. Singapore, Asia-Pacific Association of Cataract and Refractive Surgeons. Available at: [http://www.apacrs.org/barrett\\_universal2/](http://www.apacrs.org/barrett_universal2/) Accessed February 21, 2016
- F. Hill W, "The Pre-Operative Measurement of Corneal Astigmatism: Strategies for Success," the David Paton Medal

Lecture, presented at the 29th annual Cullen Course, Baylor College of Medicine, Houston, Texas, USA, February 2014



First author:

Adi Abulafia, MD

*Ein-Tal Eye Center (Abulafia, Assia),  
Tel-Aviv, Israel*